The tool is just the **hammer.** The platform is the **house.**

Build with AI speed. Govern with enterprise confidence.

The Innovative Maker is the *game changer.*

Business Innovators start in **vibe.powerapps.com** — describe an idea, ship a working app. Pro Developers ship in **VS Code** with an approved AI coding assistant and full CI/CD. In between sits the persona that breaks the old ceiling: the **Innovative Maker**, who fluidly moves between vibe.powerapps.com and VS Code — *given incredible AI superpowers, but contained and controlled by the platform.*

Business Innovators ideate · Innovative Makers cross over · Pro Developers harden · Power Platform governs every stage.

1 IDEATE & BUILD

Business Innovator

vibe.powerapps.com

- ✓ Describe the idea in natural language
- ✓ AI builds the first working app
- ✓ Validate with real users in days
- ✓ Hand off — or keep going

★ GAME CHANGER

2 CROSS-OVER · CONTAINED SUPERPOWERS

Innovative Maker

vibe.powerapps.com **and** VS Code + approved AI assistant

- ✓ Starts in vibe.powerapps.com *or* VS Code — whichever fits the task
- ✓ Pulls Code Apps into VS Code when vibe hits a ceiling — and keeps building from there
- ✓ Uses approved agent tooling to extend, refactor, and harden at AI speed
- ✓ Guardrails via `copilot-instructions.md` and `AGENTS.md`
- ✓ **Same artifact, same Dataverse, same governance — either way**

3 HARDEN & SCALE

Pro Developer

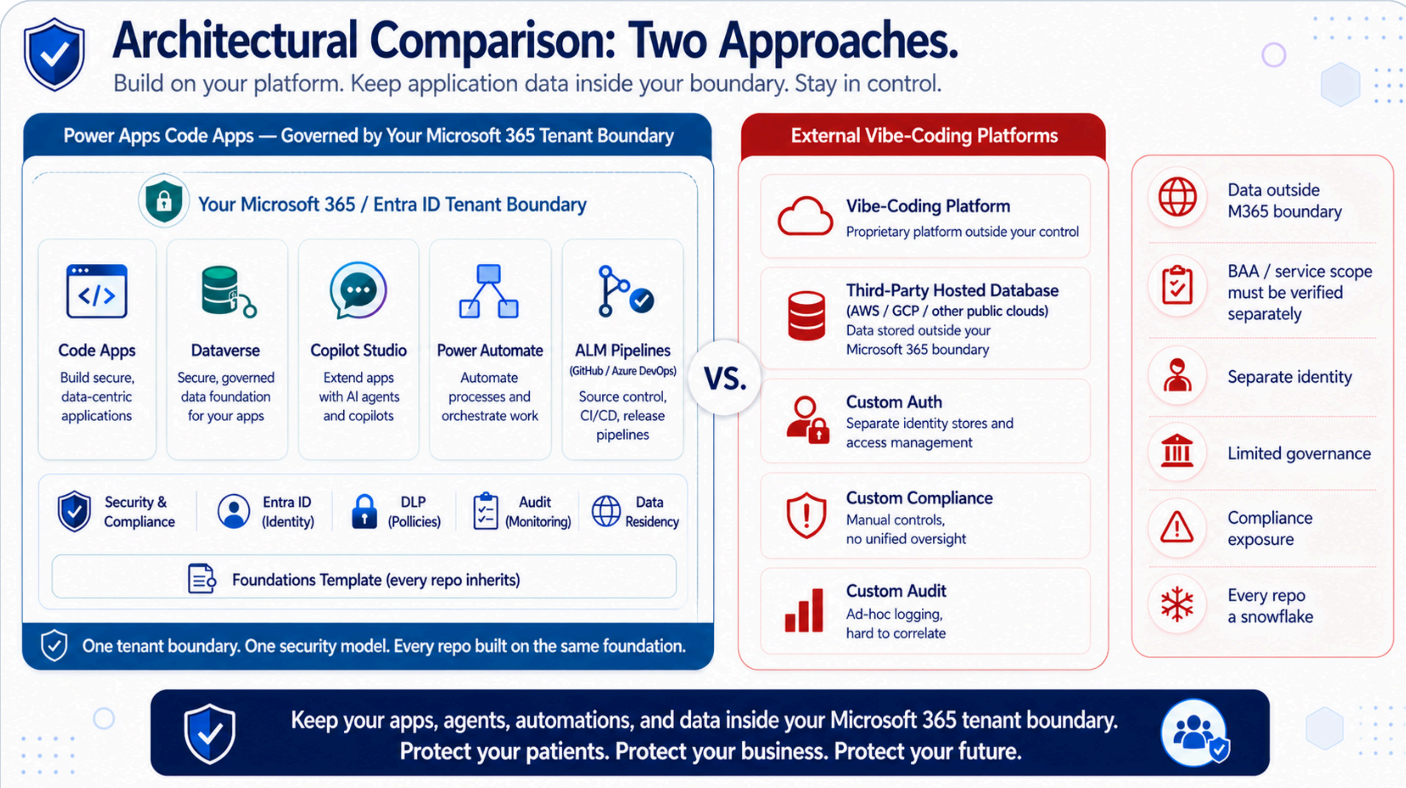
VS Code + GitHub + ALM

- ✓ Custom components, integrations, complex logic
- ✓ Source control, code review, CI/CD pipelines
- ✓ Performance, security, observability
- ✓ Ship to Managed Environments with confidence

These are roles, not necessarily different people. Often a Business Innovator *becomes* the Innovative Maker the moment vibe hits its limit — and the platform makes that step continuous instead of a cliff. The Pro Developer joins when CI/CD, custom components, or hardening matter. Same solution, same Dataverse, same governance throughout **— that's what makes the cross-over safe.**

Configure once. Let 1,000 makers self-serve.

Admins configure ALM pipelines and **Managed Environments** once. After that, makers ship to dev → test → prod with proper promotion, DLP, and approval gates baked in.



Architectural comparison — build on your platform, keep your data inside your tenant boundary, stay in control.

Code Apps + Managed Environments

- One-time admin setup of pipelines, DLP, and environment groups
- Makers self-serve deployments — guardrails enforced automatically
- **Every repo inherits the same governed foundation** — instructions, ALM, security, tests, agent guardrails — from a versioned template (see Chapter 10)
- Update the foundation once → every downstream repo can pull it in
- Solution packaging carries data model + UI + logic together
- Tenant-wide audit, telemetry, and CoE visibility across the fleet

External Vibe Platforms — Repos Without a Foundation

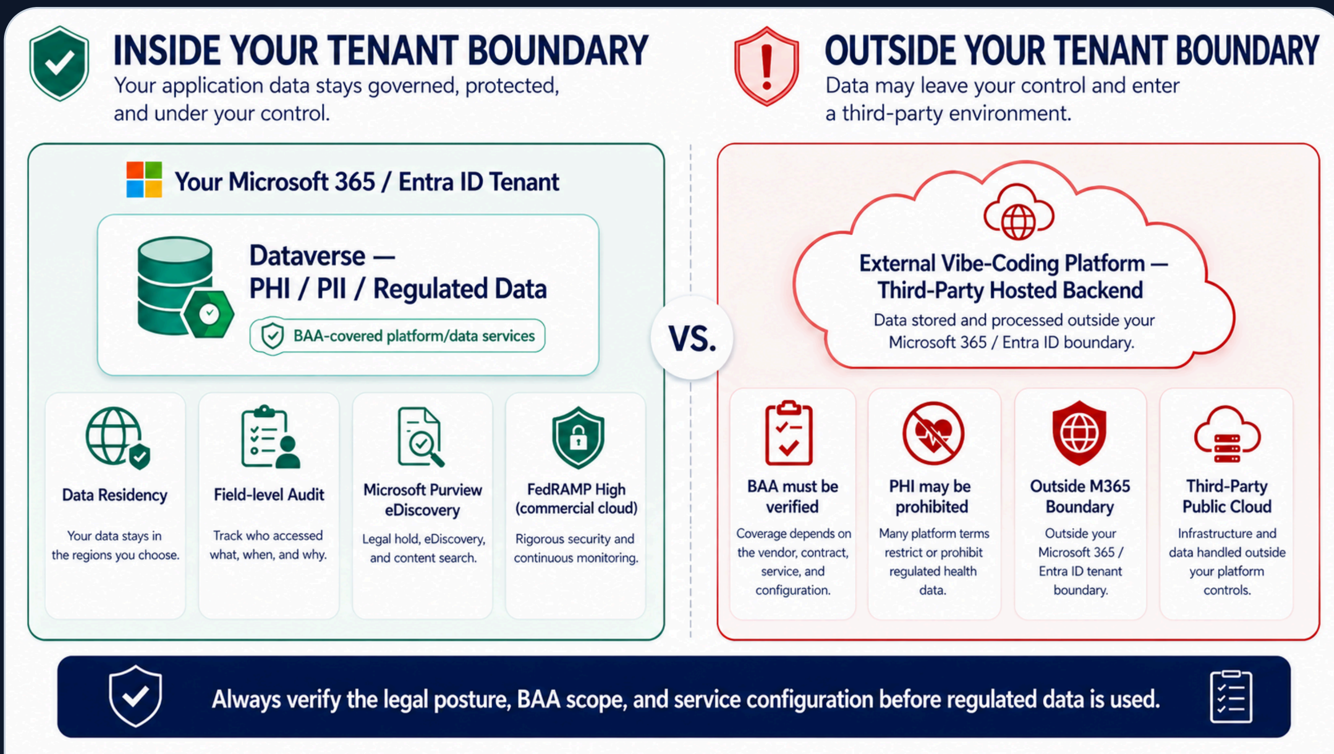
- **Every repo starts from scratch** — no inherited instructions, no inherited ALM, no inherited tests
- Governance must be re-invented per repo, per team
- Each external data backend (hosted Postgres, Firebase, Supabase, etc.) is its own snowflake
- No tenant-wide DLP, no native solution packaging, no way to push a fix across the fleet
- **Estimated +1.5 to +2.5 FTEs** of platform/governance overhead at scale (illustrative, varies by org)

The problem isn't repo count — Code Apps create plenty of repos too. The problem is repos that don't inherit a common foundation.

⚠ CHAPTER 03 · THE BOUNDARY

This is not a feature comparison. It is a boundary.

For any organization handling PHI, this single chapter changes the conversation. It's a contractual and regulatory boundary — not a preference.



Data must stay inside your tenant boundary — protect patient data, preserve compliance, maintain trust.

✓ Inside the Fence

CODE APPS WITH DATAVERSE

- ✓ Microsoft-operated SaaS governed by your **Microsoft 365 / Entra ID tenant boundary**, with regional data residency
- ✓ Covered by a **signed Microsoft BAA**
- ✓ PHI, PII, and regulated workloads explicitly supported
- ✓ FedRAMP High and HITRUST attestations apply on the Microsoft commercial cloud; sovereign-cloud options (GCC, GCC High, DoD) available — verify Code Apps availability in your target cloud
- ✓ Field-level audit and change tracking native to Dataverse; integrated with Microsoft Purview eDiscovery

Outside the Boundary

EXTERNAL VIBE-CODING PLATFORMS

- ✕ Default data layer is a **third-party hosted backend** (hosted Postgres, Firebase, Supabase, etc.) on a public cloud you don't control — outside your Microsoft 365 tenant boundary
- ✕ No Microsoft BAA coverage
- ✕ Many of these platforms' **own published terms explicitly prohibit PHI** — see proof drawer below for one transparent example
- ✕ Cannot be easily relocated into a customer-controlled Microsoft cloud boundary

SCOPE NOTE

This boundary describes the **application platform and data layer**. Coding assistants are selected separately: teams handling regulated data should route prompts only through AI services their legal and security teams have approved for that workload.

This pattern is common across the category. We use **Lovable** here as a representative example because their terms are particularly explicit. Replit, Bolt, v0, and similar platforms typically carry comparable restrictions in their own published terms — always verify the current legal documents of any platform you evaluate.

TERMS & CONDITIONS lovable.dev/terms (https://lovable.dev/terms)

*"You agree not to upload, input, or otherwise provide any **protected health information under HIPAA**, or any other sensitive categories of data... Our Services are **not designed to handle that type of data**, and we disclaim all responsibility if you choose to submit it."*

DATA PROCESSING AGREEMENT · §8

lovable.dev/data-processing-agreement (https://lovable.dev/data-processing-agreement)

*"The Customer shall **not provide any data to Lovable which is classified as sensitive**. For the avoidance of doubt, the Customer agrees not to upload... any **protected health information under HIPAA**..."*

PRIVACY POLICY lovable.dev/privacy (https://lovable.dev/privacy)

Explicitly warns against uploading sensitive health data.

KEY TAKEAWAY FOR HEALTHCARE & LIFE SCIENCES

Microsoft signs a BAA and keeps your data inside your Microsoft 365 tenant boundary. The **external vibe-coding category, broadly, does not** — and many of these platforms explicitly prohibit PHI in their own published terms. Always confirm the legal posture of any platform before letting regulated data near it.

CHAPTER 04 · AGENTIC AI

One intelligent connection for agents.

In the era of intelligent agents, the data layer must speak the agent's language. Many databases now ship MCP servers — including Azure SQL, Cosmos DB, and Postgres — but they expose *schema*. **Dataverse's MCP exposes business semantics:** security roles, business units, business rules, choice sets, relationships, and audit — the things agents actually need to act safely on enterprise data.

DATAVERSE

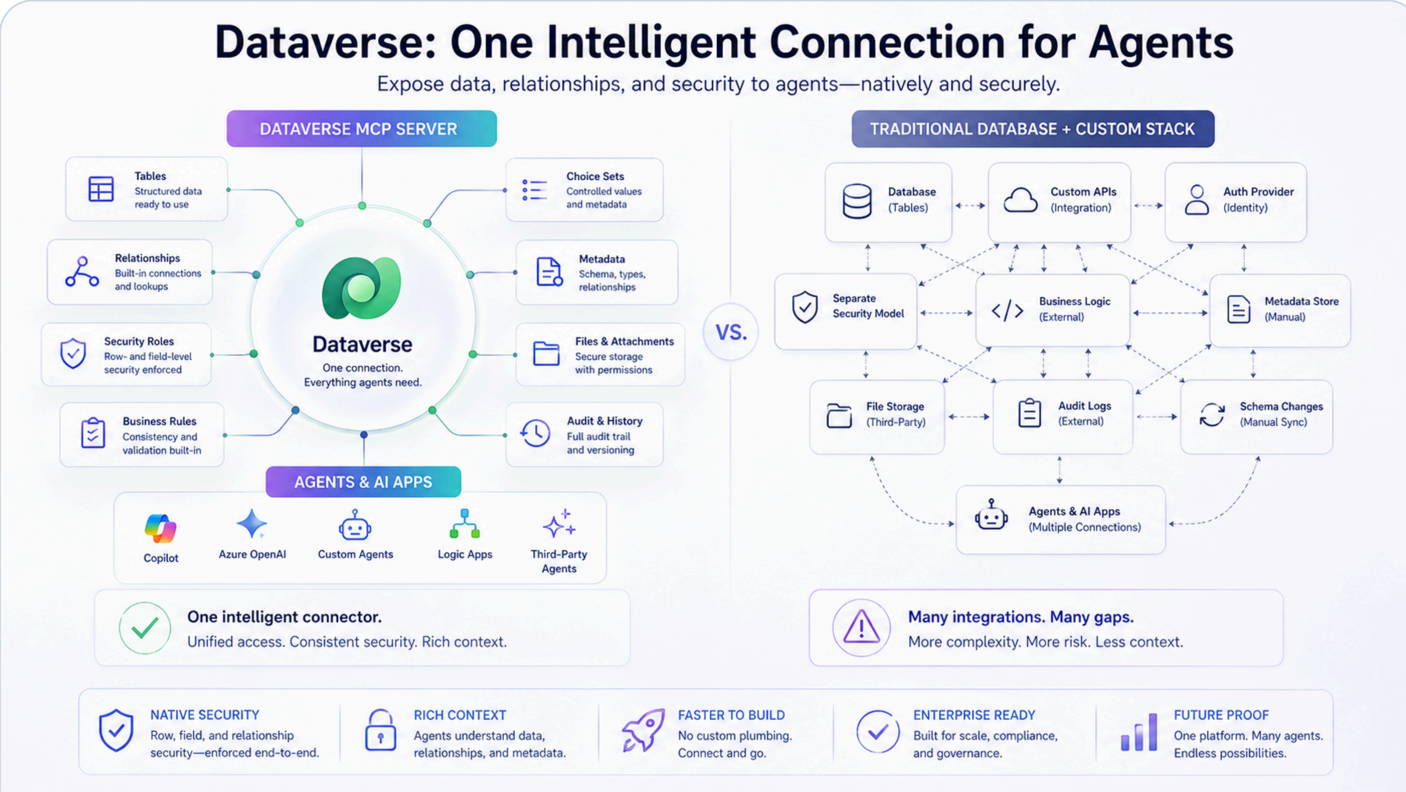
Total context. One connection.

- ✓ Auto-discover tables, columns, relationships
- ✓ Row & column security flow through
- ✓ Business rules & choice sets introspectable
- ✓ Copilot Studio, M365 Copilot, custom agents

SCHEMA-ONLY MCP

Plumbing without meaning.

- ✕ MCP exposes columns and types, not business rules or security roles
- ✕ Row/column security must be re-described to every agent
- ✕ Constant drift between schema, prompts, and policy
- ✕ Each new agent restarts the integration cost

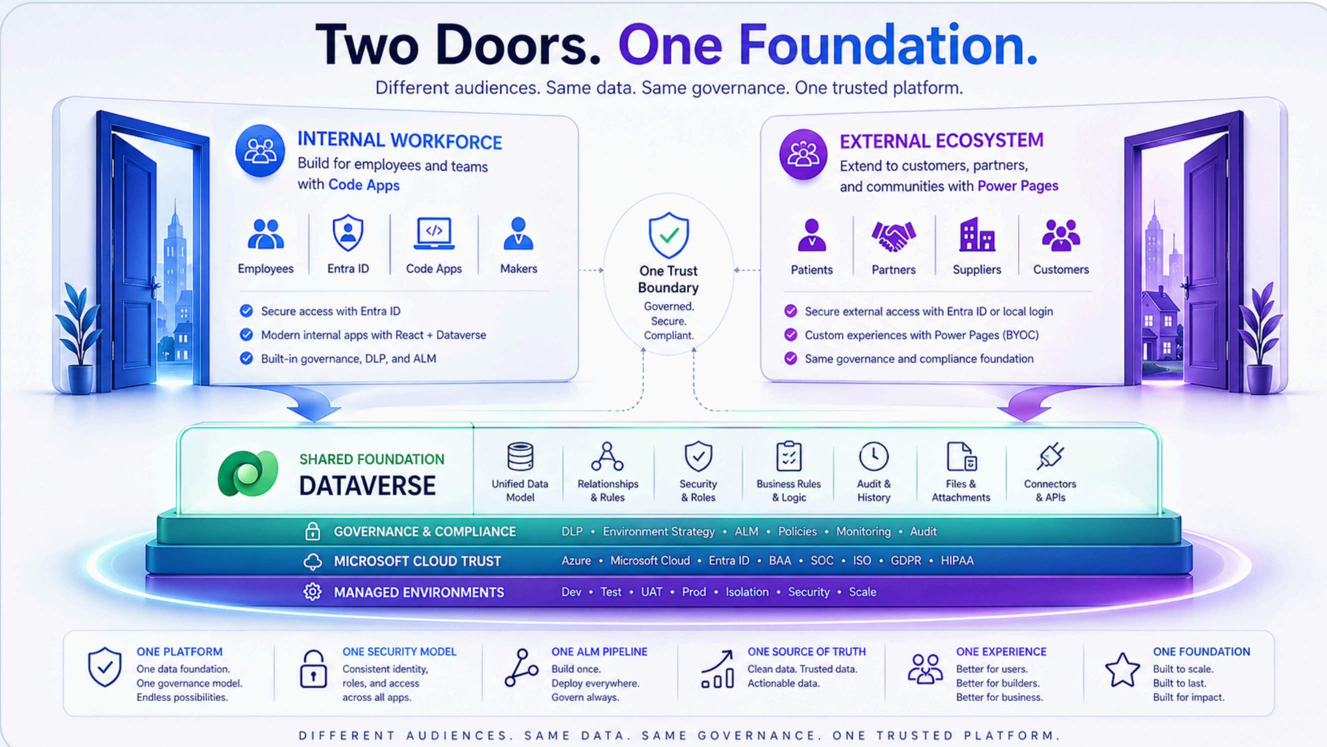


Dataverse exposes tables, relationships, metadata, security, and business rules — natively to agents.

CHAPTER 05 · INTERNAL & EXTERNAL

Two Doors. One Foundation.

Different audiences. Same data. Same governance. One trusted platform.



⌂ Internal Workforce

CODE APPS · ENTRA ID

- ✓ Modern internal apps with React + Dataverse
- ✓ Secure access with Entra ID
- ✓ Built-in governance, DLP, and ALM

External Ecosystem

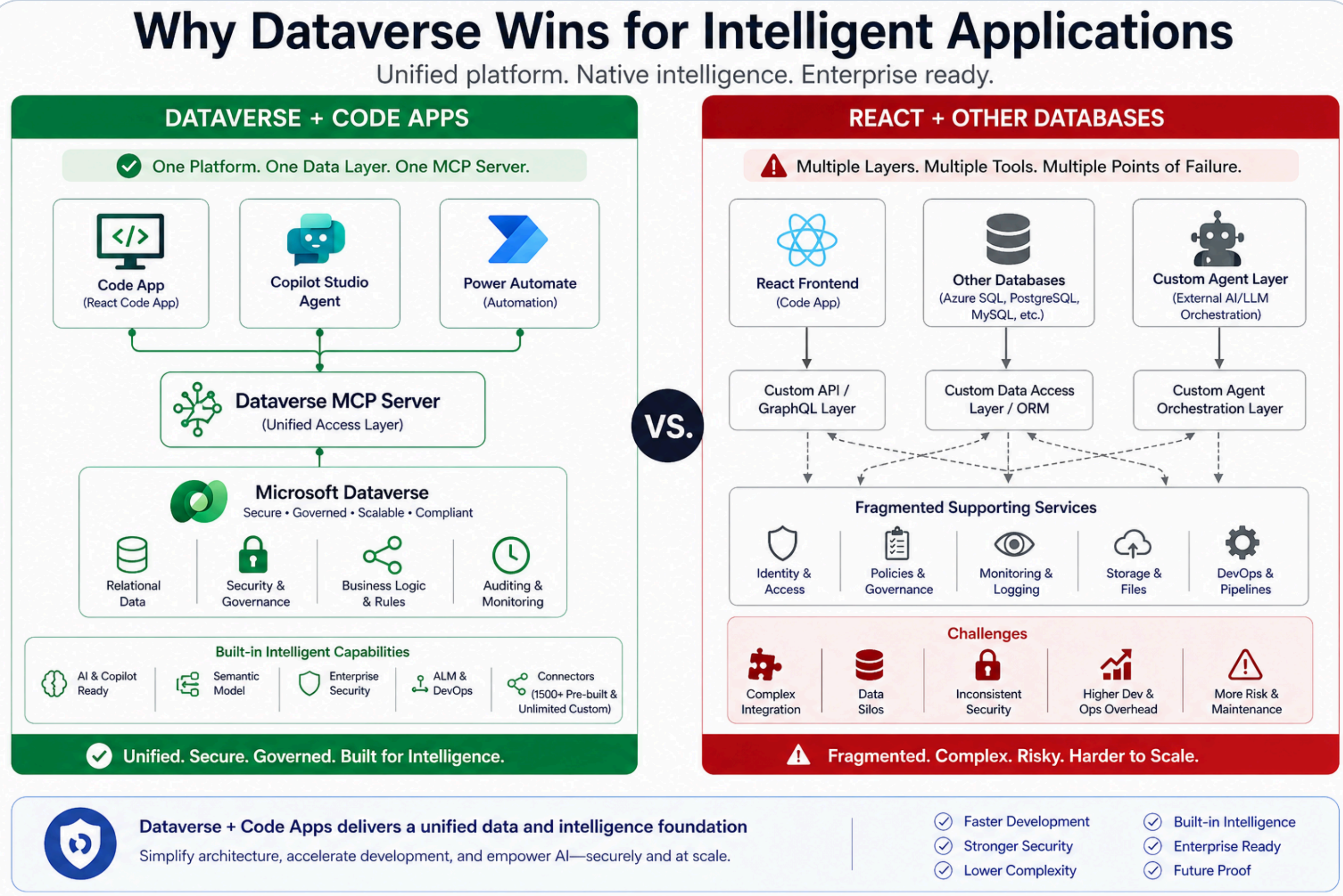
POWER PAGES · BYOC

- ✓ Customers, patients, partners, suppliers
- ✓ Modern React on the secure Power Pages runtime
- ✓ Same Dataverse security, end-to-end

ONE PLATFORM One data foundation	**ONE SECURITY** Identity, roles, access
ONE ALM Build once. Govern always.	**ONE TRUTH** Clean, trusted data
ONE EXPERIENCE Users · Builders · Business	**ONE FOUNDATION** Built to scale & last

A traditional database is just a database. Dataverse is a platform.

Dataverse is a governed business application platform. The difference shows up in nearly every dimension that matters at enterprise scale.



Why Dataverse wins for intelligent applications — unified platform, native intelligence, enterprise ready.

CAPABILITY	DATAVERSE	OTHER DATABASES
Row & column security	✓ Built-in security roles, business units, hierarchical, field-level	✕ Custom-built with views, RLS policies, and app-tier code
MCP Server	✓ Native — exposes business semantics: security roles, business	✕ Where MCP exists, exposes *schema only* — security and

	rules, choice sets, relationships, audit	business rules must be re-described per agent
Agent intelligence	✓ Copilot Studio + M365 Copilot speak Dataverse natively	✗ Requires bespoke prompt engineering and middleware
Business logic location	✓ Centralized: business rules, plug-ins, flows	✗ Scattered across app code, stored procs, triggers, services
ALM & solution packaging	✓ First-class solutions carry schema + UI + logic together	✗ DACPAC + custom scripts + bespoke release pipelines

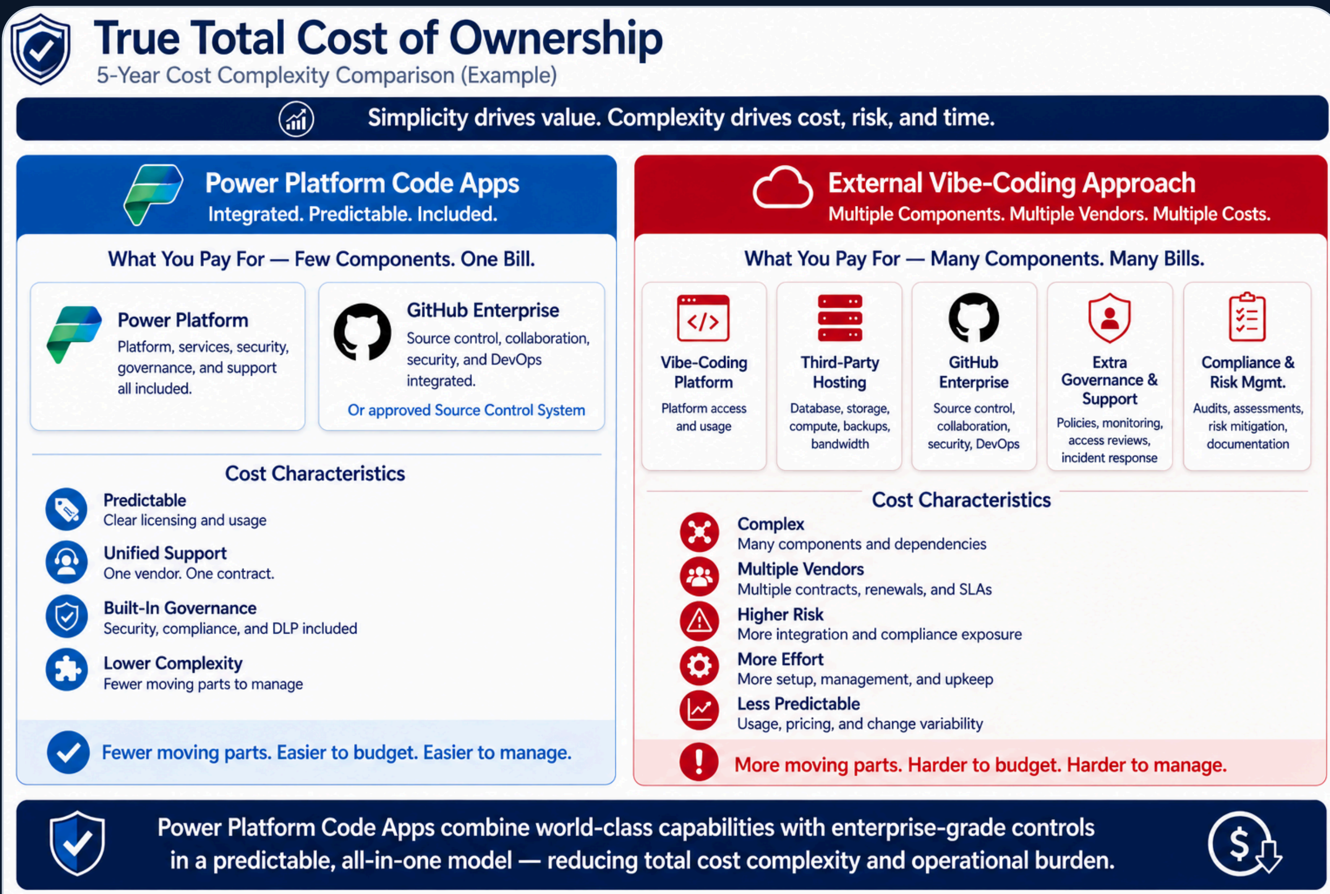
Show 4 more rows ▾

CHAPTER 07 · TRUE TOTAL COST OF OWNERSHIP

The hidden cost of tool sprawl.

Simplicity drives value. Complexity drives cost, risk, and time.

We deliberately don't put dollar signs here — pricing varies by scale, region, and deal. The point is the **shape** of the cost structure.



True total cost of ownership — simplicity drives value, complexity drives cost, risk, and time.

POWER PLATFORM CODE APPS

Integrated. Predictable. Included.

What you pay for: Few components. One bill.

PP **Power Platform**
Platform, services, security, governance, support — all included

GH **GitHub Enterprise**
Source control, collaboration, security, DevOps

Predictable Clear licensing	**Unified Support** One vendor
Built-in Governance Compliance & ALM	**Future-Compatible** Fewer parts

EXTERNAL VIBE PLATFORM APPROACH

Many components. Many bills.

What you pay for: Many components. Many bills.

- **VP** **Vibe-Coding Platform**
  Subscription, usage, compute, backups
- **DB** **Third-Party Data Backend**
  Hosted database on AWS / GCP / other public clouds — storage, networking
- **GH** **GitHub Enterprise**
  Repos, security, governance, DevOps
- **+G** **Extra Governance & Support**
  Policies, monitoring, incident response, added FTEs
- **C&R** **Compliance & Risk Mgmt**
  Audit, logging, certifications, documentation

Complex Multiple bills	**Higher Risk** More integration surface
Reduced Reliability Less consistency	**Higher Maintenance** Vendor sprawl

BOTTOM LINE

One platform. Fewer moving parts. Greater value.

CHAPTER 08 · AI FLEXIBILITY

Bring the right frontier model. Always know which one you're using.

Coding-model leadership shifts month to month. The **VS Code ecosystem** lets developers use the assistant route their organization approves — GitHub Copilot, Claude Code, Codex, Cursor, Cline, or direct API integrations — while still building the same governed Code App on Dataverse. The point is not one assistant. It is transparent routing, named models, and a platform foundation that remains consistent as the AI layer changes.

Below: a cross-section of leading coding models and assistant surfaces available across the VS Code ecosystem in 2026. Availability, contractual coverage, and data-handling posture vary by route; regulated teams should validate the approved path with legal, security, and procurement.

ANTHROPIC

Claude Opus 4.x

Sonnet 4.x · Haiku 4.5

Industry-leading agentic coding & long-horizon refactors.

OPENAI

GPT-5 / GPT-5 Codex

o-series reasoning

Strong general coding & deep reasoning workflows.

GOOGLE

Gemini 2.5 Pro

Gemini Code Assist

Massive context windows for whole-repo reasoning.

GITHUB

GitHub Copilot

Multi-model: Claude · GPT · Gemini

A productive default for many workloads; confirm fit for regulated data-handling scenarios.

CURSOR

Cursor + Composer

Routes to Claude / GPT / Gemini

IDE-native agent loops, model of your choice.

XAI

Grok Code

Grok 4 family

Real-time web context for current API knowledge.

DEEPSEEK

DeepSeek-Coder V3

Open-weights option

Strong code completion at lower cost-per-token.

ALIBABA

Qwen Coder

Open-weights, multilingual

Self-hostable for sovereign workloads.

The cadence is the point: new frontier coding models drop every few weeks. Within **days to weeks** of release, they typically appear in at least one VS Code-compatible surface — GitHub Copilot, Cursor, Cline, or direct API integrations. Yesterday's leader rarely stays the leader for long — and your developers will always know which one they're running.

GitHub Copilot

A strong default for many developer workflows. Treat it as one approved option, not the only AI path.

Claude Code via Azure AI Foundry

For Claude-based agent work, route the CLI or VS Code extension to an approved Foundry-hosted endpoint where available.

Codex via Azure OpenAI

For OpenAI coding agents, route through your Azure OpenAI resource using Entra ID or managed keys under your Azure controls.

The practical pattern is simple: keep the **application, data model, ALM, and governance** in Code Apps + Dataverse, then let legal and security approve which AI assistant endpoint can receive prompts for each workload. For PHI-adjacent scenarios, confirm the applicable BAA and service scope in writing before enabling that route.

Code Apps — Open & Transparent

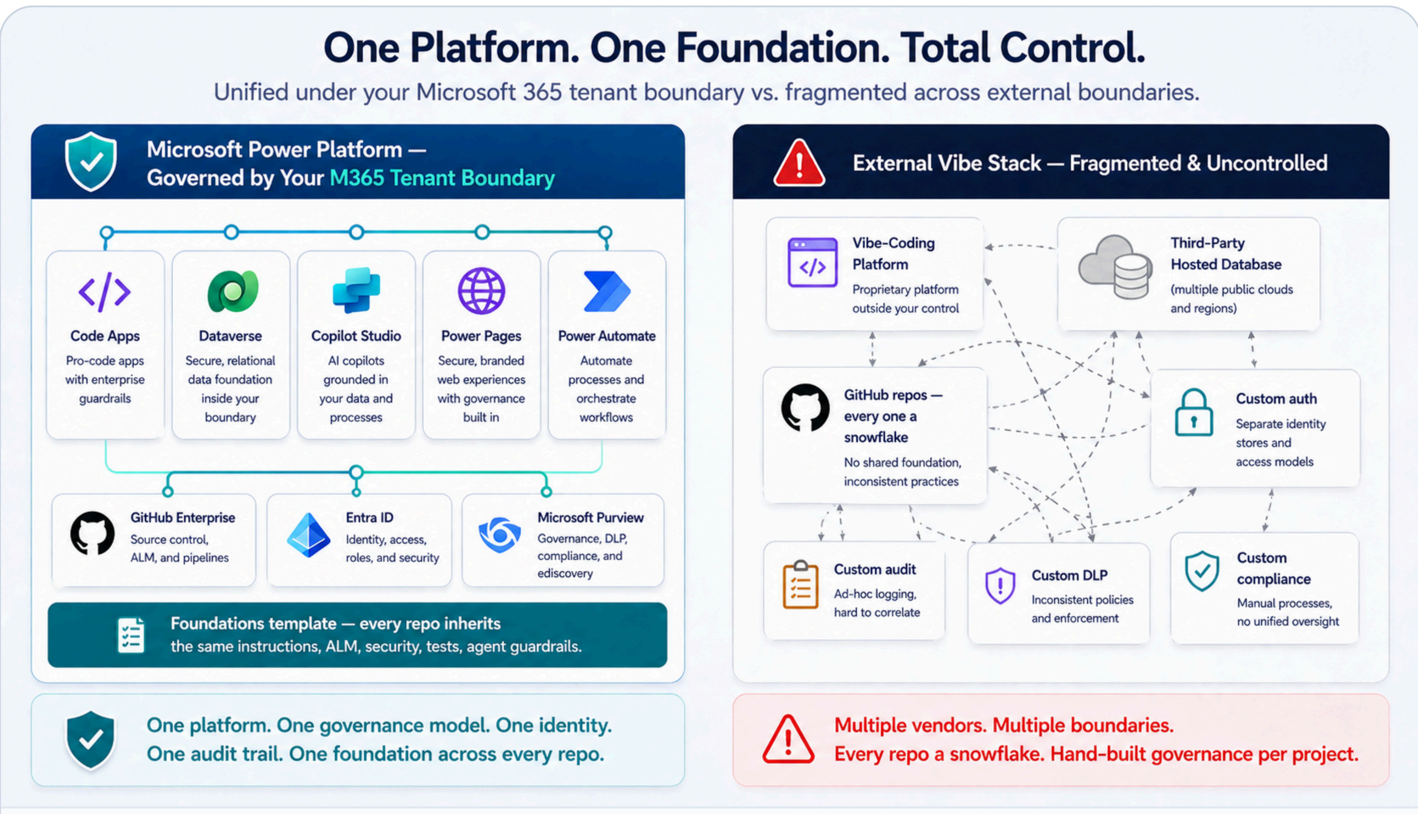
- ✓ **Named, versioned models** — you always know what wrote your code
- ✓ Approved VS Code assistants work here: Copilot, Claude Code, Codex, Cursor, Cline, or direct APIs
- ✓ Switch per repo, per developer, per task — based on both capability and compliance posture
- ✓ Enterprise controls depend on the approved route: GitHub Enterprise, Azure OpenAI, Azure AI Foundry, or other governed endpoints
- ✓ `copilot-instructions.md` sets guardrails across every assistant
- ✓ New frontier model today → available within days

External Vibe Platforms — Opaque Models

- ✕ **No model transparency** — you don't know which model or version
- ✕ Cannot verify whether you're on the latest frontier capability
- ✕ The platform vendor decides the model roster — switching is not yours
- ✕ No version history, no audit trail of which model produced what
- ✕ Enterprise key management lives outside customer control
- ✕ In a landscape that shifts weekly, opacity is a liability

CHAPTER 09 · THE BIGGER PICTURE

Fragmentation feels fast at first. Platform coherence wins at scale.



Unified platform vs fragmented tools — connected, governed, reliable, built to scale.

CHAPTER 10 · FOUNDATIONS

Getting started is the hardest part — until now.

Most teams spend their first month discovering platform quirks the hard way. **PAppsCAFoundations** encodes every one of those lessons into a repeatable, tested path — so your team builds features on day one, not debugs platform quirks.

PAppsCAFoundations — a proven foundation, a clear path, faster to value.

🧭

Consulting-Grade Methodology

Business decomposition → scope refinement → Dataverse planning → prototype validation → build & deploy. Encoded in your repo.

🤖

Agent-Guided Development

14 scoped instruction files plus `AGENTS.md` — Copilot, Cursor, Claude Code, every agent follows the same rules.

⚡

Interactive Setup Wizard

A 9-step wizard verifies prerequisites, scaffolds React + Fluent UI + Dataverse, seeds prototype assets, runs smoke tests.

🧪

Testable from Day One

Vitest, Playwright, MSW ship in the scaffold. Smoke tests pass on first run. Mocks let you test before Dataverse exists.

🔐

Security-First Auth

1Password CLI integration or AES-256-GCM encrypted secrets. Pre-commit hooks block accidental secret commits.

🔄

Foundations Sync

Pull the latest instructions, scripts, and wizard into your downstream project. No fork management. No merge conflicts.

OPEN SOURCE · MIT LICENSED

Start strong. Build smart. Deliver impact.

Clone the template, run the wizard, start building. Your team gets the accumulated lessons of every team that came before.

Open Repo → (https://github.com/martycarreras-psnl/PAppsCAFoundations)

See Setup Path (https://martycarreras-psnl.github.io/PAppsCAFoundations/guide.html)

Start Building (https://aka.ms/codeappsfoundations)

CHAPTER 11 · THE VERDICT

If you're building serious enterprise applications, platforms win.

Build with AI speed. **Govern with enterprise confidence.**

Choose Code Apps when you value:

- ✓ Tenant-resident, BAA-covered data (PHI, PII, regulated)
- ✓ A maturity model from citizen to pro developer
- ✓ Native MCP and agent intelligence
- ✓ Configure-once governance for 1,000+ makers
- ✓ External-facing apps via Power Pages BYOC
- ✓ Freedom to use any frontier AI model in VS Code

Reconsider external vibe platforms when:

- ✕ PHI or other regulated data is in scope
- ✕ You need tenant-wide governance and ALM
- ✕ Repos that don't inherit a common foundation become unmanageable
- ✕ Agent integration is a strategic priority
- ✕ Long-term maintainability matters more than first-week velocity

"The AI coding tool is just the hammer. The platform is the house."

For organizations that value governance, compliance, agent integration, long-term maintainability, and the ability to scale citizen development safely — **especially when PHI is involved** — Power Apps Code Apps with Dataverse is the clear, defensible enterprise choice.

Start with Foundations → (https://martycarreras-psnl.github.io/PAppsCAFoundations/)

Back to Top

DISCLAIMER

This content is **not a formal statement, position, or endorsement from Microsoft**. It is an observational perspective from **Marty Carreras** (https://www.linkedin.com/in/carrema/), Microsoft's US Healthcare & Life Sciences AI Business Soltions CTO, offered to help organizations think through enterprise application strategy. Opinions are the author's own.

Power Apps Code Apps Advantage · Build with AI speed. Govern with enterprise confidence.